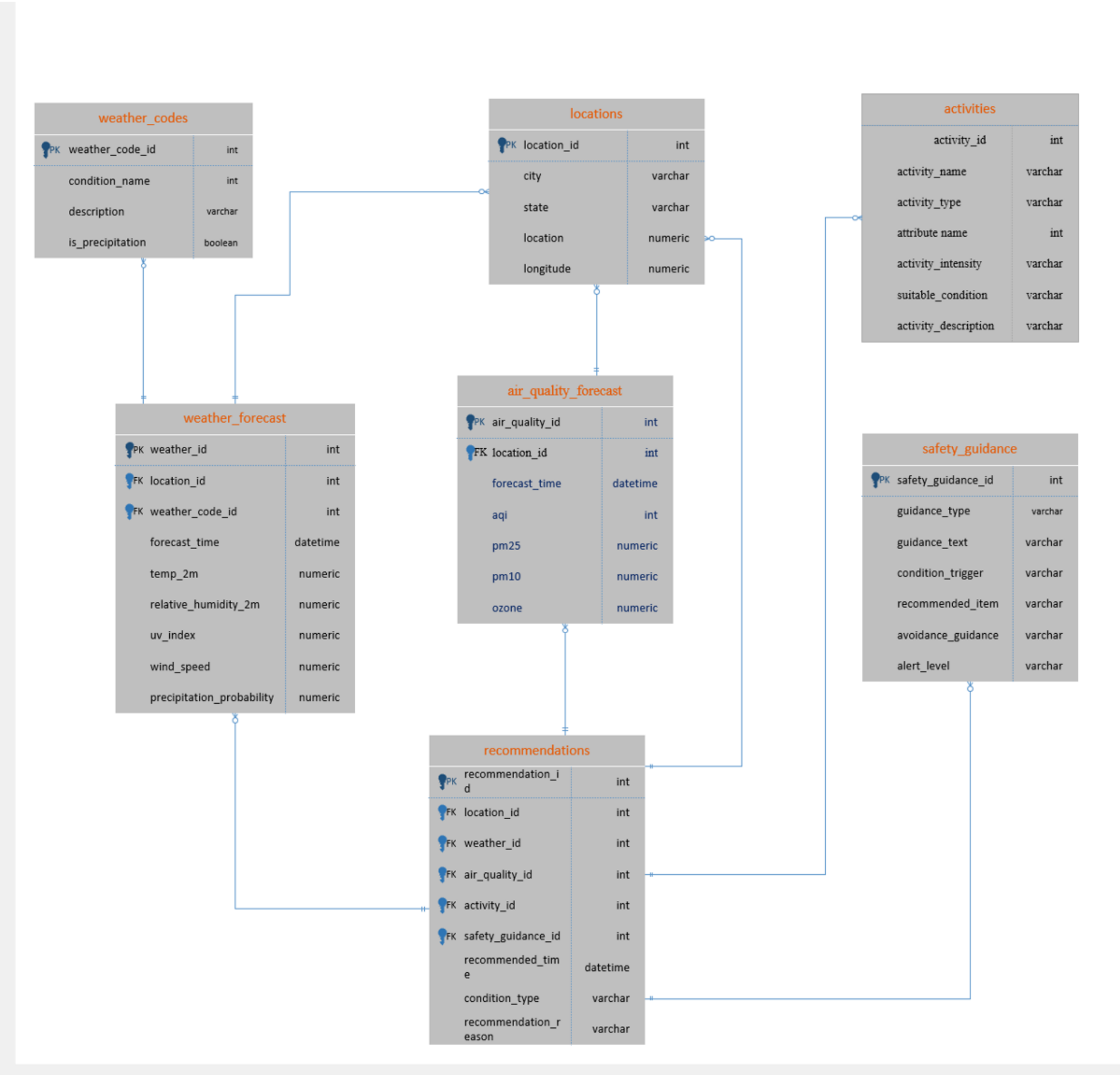
Database Schema Documentation
Environmental Activity & Safety Recommendation Data Warehouse

This database stores environmental forecast data, air quality forecast data, activity recommendation data, and environmental safety guidance loaded into a PostgreSQL database. The database supports a Python ETL pipeline, normalized storage, and Power BI dashboard reporting for the Environmental Activity & Safety Recommendation Dashboard.

The schema is normalized to approximately Third Normal Form (3NF) by:

- separating entities into related tables
- avoiding repeated descriptive data
- storing reusable weather codes, activities, and safety guidance in reference tables
- using foreign keys to connect forecast data to recommendation outputs
- using the recommendations table as a recommendation output bridge connecting forecasts, activities, and safety guidance

ER Diagram
Environmental Activity & Safety Recommendation ER Diagram



Database Overview

The database contains seven tables:

- weather_codes
- weather_forecast
- locations
- air_quality_forecast
- activities
- safety_guidance
- recommendations

Entity Relationship Summary

Table	Purpose

weather_codes	Stores standardized weather condition codes, labels, and descriptions used by weather forecasts.
weather_forecast	Stores forecasted weather measurements from the Open-Meteo Weather API.
locations	Stores supported city and coordinate information for API requests and reporting.
air_quality_forecast	Stores air quality forecast measurements from the Open-Meteo Air Quality API.
activities	Stores indoor and outdoor activity options used by the recommendation engine.
safety_guidance	Stores reusable safety guidance such as clothing, sunscreen, umbrellas, hydration, and exposure warnings.
recommendations	Bridge/output table connecting weather, air quality, activity, and safety guidance records into final dashboard recommendations.

Table Documentation

1. weather_codes

Purpose
Stores standardized weather condition labels and descriptions used by the recommendation engine. This table prevents repeated weather condition descriptions from being stored in every forecast record.

Example
- Code 1 = Clear Sky
- Code 7 = Light Rain
- Code 8 = Heavy Rain
- Code 10 = Heavy Snow
- Code 12 = Poor Air Quality

Primary Key
- weather_code_id

Relationships
- One weather code can relate to many weather_forecast records.
- Referenced by weather_forecast.weather_code_id.

Table Structure

Column Name	Data Type	Key	Description
weather_code_id	INTEGER	Primary Key	Unique weather condition identifier used by weather_forecast.
condition_name	VARCHAR		Weather condition name such as Clear Sky, Heavy Rain, Snow, Heat, Windy, or Humid.
description	VARCHAR		Human-readable explanation of the weather condition.
is_precipitation	BOOLEAN		Indicates whether the condition is related to rain or snow.

2. weather_forecast

Purpose
Stores weather forecast information retrieved from the Open-Meteo Weather API.

Details

- Forecast data includes temperature, humidity, UV index, wind speed, and precipitation probability.

Primary Key

- weather_id

Foreign Keys

- location_id -> locations.location_id
- weather_code_id -> weather_codes.weather_code_id

Relationships

- Many forecasts can reference one weather code.
- Many forecasts can belong to one location.
- One forecast can support one or more recommendation outputs.

Table Structure

Column Name	Data Type	Key	Description
weather_id	INTEGER	Primary Key	Unique weather forecast record ID.
location_id	INTEGER	Foreign Key	Location associated with the forecast.
weather_code_id	INTEGER	Foreign Key	Weather code associated with the forecast.
forecast_time	DATETIME		Date and time of the forecast record.
temperature_2m	NUMERIC		Temperature at 2 meters.
relative_humidity_2m	NUMERIC		Relative humidity percentage at 2 meters.
uv_index	NUMERIC		UV exposure index used for sun protection guidance.
wind_speed_10m	NUMERIC		Wind speed at 10 meters.
precipitation_probability	NUMERIC		Probability of rain or snow.

3. locations

Purpose

Stores unique supported locations and geographic coordinates used for Open-Meteo API requests and dashboard filtering.

Primary Key

- location_id

Relationships

- One location can have many weather forecast records.
- One location can have many air quality forecast records.
- One location can generate many recommendation outputs.

Constraints

- The combination of city, state, latitude, and longitude should uniquely identify a supported location.

Table Structure

Column Name	Data Type	Key	Description
location_id	INTEGER	Primary Key	Unique location identifier.

city	VARCHAR		Supported city name (Louisville)
state	VARCHAR		State abbreviation or state name (KY)
latitude	NUMERIC		Latitude used for Open-Meteo API requests.
longitude	NUMERIC		Longitude used for Open-Meteo API requests.

4. air_quality_forecast

Purpose
Stores air quality forecast information retrieved from the Open-Meteo Air Quality API.

Details
- Forecast data includes AQI, PM2.5, PM10, and ozone measurements.

Primary Key
- air_quality_id

Foreign Keys
- location_id -> locations.location_id

Relationships
- Many air quality forecasts can belong to one location.
- One air quality forecast can support one or more recommendation outputs.

Table Structure

Column Name	Data Type	Key	Description
air_quality_id	INTEGER	Primary Key	Unique air quality forecast record ID.
location_id	INTEGER	Foreign Key	Location associated with the air quality forecast.
forecast_time	DATETIME		Date and time of the air quality forecast record.
aqi	INTEGER		Air Quality Index value.
pm25	NUMERIC		Fine particulate matter measurement.
pm10	NUMERIC		Larger particulate matter measurement.
ozone	NUMERIC		Ozone concentration measurement.

5. activities

Purpose
Stores indoor and outdoor activity options used by the recommendation engine.

Example
- Walking = Outdoor, Low intensity, suitable for mild weather and good AQI
- Cycling = Outdoor, Moderate intensity, suitable for low wind and dry conditions
- Museum Visit = Indoor, Low intensity, suitable for rain, snow, poor AQI, or extreme UV
- Board/Card Games at Home = Indoor, Low intensity, suitable for heavy rain, snow, or severe weather

Primary Key
- activity_id

Relationships

- One activity can be reused across many recommendation outputs.
- Referenced by recommendations. activity_id.

Table Structure

Column Name	Data Type	Key	Description
activity_id	INTEGER	Primary Key	Unique activity identifier.
activity_name	VARCHAR		Recommended activity name.
activity_type	VARCHAR		Indoor or outdoor activity classification.
activity_intensity	VARCHAR		Low, moderate, or high activity intensity.
suitable_condition	VARCHAR		Environmental condition where the activity is appropriate.
activity_description	VARCHAR		Brief explanation of why the activity may be recommended.

6. safety_guidance

Purpose

Stores reusable environmental safety guidance such as clothing recommendations, sunscreen guidance, rain protection, winter protection, hydration reminders, and exposure warnings.

Example

- High UV -> SPF 50+ sunscreen, hat, sunglasses
- Rain -> umbrella and waterproof shoes
- Cold/Snow -> winter coat, gloves, and boots
- Poor AQI -> limit prolonged outdoor exposure
- High Temperature -> water bottle and shaded activity planning

Primary Key

- safety_guidance_id

Relationships

- One safety guidance rule can be reused across many recommendation outputs.
- Referenced by recommendations. safety_guidance_id.

Table Structure

Column Name	Data Type	Key	Description
safety_guidance_id	INTEGER	Primary Key	Unique safety guidance identifier.
guidance_type	VARCHAR		Type of guidance such as Clothing, Hydration, Sun Protection, or Air Quality.
guidance_text	VARCHAR		Safety guidance message shown to the user.
condition_trigger	VARCHAR		Condition that activates the guidance.
recommended_item	VARCHAR		Recommended item such as sunscreen, umbrella, water bottle, or winter coat.
avoidance_guidance	VARCHAR		Condition or behavior the user should avoid.
alert_level	VARCHAR		Low, Moderate, or High alert level.

7. recommendations

Purpose
Stores final recommendation outputs generated by combining location, weather forecast, air quality forecast, activity options, and safety guidance.

Details
- This table acts as the recommendation output and bridge table for the project.
- It connects forecast conditions to recommended activities and safety guidance.
- It supports the Power BI dashboard recommendation cards and safety alerts.

Example
- Heavy Rain -> Museum Visit + Umbrella Guidance
- High UV -> Shaded Park Visit + SPF 50+ Guidance
- Poor AQI -> Shopping Center + Limit Outdoor Exposure

Primary Key
- recommendation_id

Foreign Keys
- location_id -> locations.location_id
- weather_id -> weather_forecast.weather_id
- air_quality_id -> air_quality_forecast.air_quality_id
- activity_id -> activities.activity_id
- safety_guidance_id -> safety_guidance.safety_guidance_id

Relationships
- Many recommendations can be generated for one location.
- Many recommendations can reference one weather forecast.
- Many recommendations can reference one air quality forecast.
- Many recommendations can reuse one activity.
- Many recommendations can reuse one safety guidance rule.

Table Structure

Column Name	Data Type	Key	Description
recommendation_id	INTEGER	Primary Key	Unique recommendation output identifier.
location_id	INTEGER	Foreign Key	Links the recommendation to a location.
weather_id	INTEGER	Foreign Key	Links the recommendation to a weather forecast record.
air_quality_id	INTEGER	Foreign Key	Links the recommendation to an air quality forecast record.
activity_id	INTEGER	Foreign Key	Links the recommendation to an activity.
safety_guidance_id	INTEGER	Foreign Key	Links the recommendation to safety guidance.
recommendation_time	DATETIME		Date and time when the recommendation was generated.
condition_type	VARCHAR		Main environmental condition driving

			the recommendation.
recommendation_reason	VARCHAR		Explanation of why the recommendation was generated.

Cardinality Relationships

Parent Table	Child Table	Relationship Type
weather_codes	weather_forecast	One-to-Many
locations	weather_forecast	One-to-Many
locations	air_quality_forecast	One-to-Many
locations	recommendations	One-to-Many
weather_forecast	recommendations	One-to-Many
air_quality_forecast	recommendations	One-to-Many
activities	recommendations	One-to-Many
safety_guidance	recommendations	One-to-Many

Bridge and Output Table Explanation

The recommendations table functions as the central recommendation output table. It is not a simple lookup table. Instead, it bridges forecast data, air quality data, activity options, and safety guidance into one final recommendation. This design allows the same activity or safety guidance rule to be reused across many different environmental conditions.

Normalization Notes (3NF)

- Weather condition descriptions are stored once in weather_codes instead of repeated in weather_forecast.
- Activities are stored once in activities instead of repeated in recommendations.
- Safety guidance is stored once in safety_guidance instead of repeated in recommendations.
- Forecast measurements are stored separately in weather_forecast and air_quality_forecast.
- The recommendations table connects records through foreign keys and stores the final recommendation output.
- Non-key columns depend only on each table primary key.

Data Sources

Weather and Air Quality Data Source

Weather and air quality data are sourced from the Open-Meteo APIs. The APIs provide:

- weather forecast data
- weather condition indicators
- temperature data
- humidity data
- UV index data
- wind speed data
- precipitation probability data
- AQI data
- PM2.5 and PM10 data
- ozone measurements

Recommendation Reference Data Source
Activity and safety guidance reference data are sourced from project reference tables prepared in Excel and loaded into PostgreSQL.

Reference Data Examples from Project Excel Files

weather_codes reference examples

weather_code_id	condition_name	description	is_precipitation
1	Clear Sky	Clear conditions suitable for outdoor activity when AQI and UV levels are acceptable.	False
2	Partly Cloudy	Mixed sun and clouds with generally flexible outdoor planning conditions.	False
3	Humid	High humidity conditions that may increase discomfort and require hydration/light clothing.	False
4	Heat	High temperature conditions requiring hydration, light clothing, and shaded activity planning.	False
5	High UV	Elevated UV exposure requiring sunscreen and sun protection.	False
6	Windy	Wind conditions that may affect outdoor comfort and require caution in exposed areas.	False
7	Light Rain	Rain conditions where covered outdoor activities or umbrella use may be appropriate.	True
8	Heavy Rain	High rain risk with wet/slippery surfaces; indoor activities are recommended.	True
9	Light Snow	Light snow conditions requiring winter clothing and caution on surfaces.	True
10	Heavy Snow	Snow conditions with reduced visibility or slippery surfaces; indoor options are preferred.	True
11	Cold Weather	Low temperature conditions requiring warm layered clothing and limited exposure.	False
12	Poor Air Quality	Elevated air pollution conditions where prolonged outdoor exposure should be reduced.	False

activities reference examples

activity_id	activity_name	activity_type	activity_intensity	suitable_condition	description
1	Walking	Outdoor	Low	Mild weather, low wind, good AQI	Light outdoor activity suitable for favorable environmental conditions.
2	Cycling	Outdoor	Moderate	Good AQI, low wind, dry conditions	Outdoor activity suitable when wind and precipitation risk are low.
3	Hiking	Outdoor	Moderate	Good AQI, mild temperature, low precipitation	Outdoor activity recommended during clear and comfortable

					conditions.
4	Swimming	Outdoor	Moderate	High temperature, low storm risk	Recommended during hot weather with hydration and sun protection.
5	Shaded Park Visit	Outdoor	Low	High UV or high temperature	Outdoor option with reduced direct sun exposure.
6	Winter Park Walk	Outdoor	Low	Cold or light snow conditions	Short outdoor activity with winter clothing and surface caution.
7	Museum Visit	Indoor	Low	Rain, snow, poor AQI, extreme UV	Indoor activity suitable during unfavorable outdoor conditions.
8	Shopping Center	Indoor	Low	Rain, snow, poor AQI	Indoor activity option when outdoor exposure should be limited.
9	Indoor Fitness	Indoor	Moderate	Cold, snow, heat, high UV, poor AQI	Indoor physical activity option during unsafe outdoor conditions.
10	Restaurant Visit	Indoor	Low	Rain, snow, extreme heat, poor AQI	Indoor social activity suitable during weather or air quality concerns.
11	Board/Card Games at Home	Indoor	Low	Heavy rain, snow, severe weather	At-home activity recommended during high precipitation or unsafe travel conditions.
12	Indoor Recreation	Indoor	Low	Windy, rainy, snowy, or poor AQI	General indoor option for days with environmental safety concerns.

safety_guidance reference examples

safety_guidance_id	guidance_type	guidance_text	condition_trigger	recommended_item	avoidance_guidance	alert_level
1	Clothing	Wear light breathable clothing.	Humid	Light clothing	Avoid prolonged heat exposure.	Moderate

2	Hydration	Carry water and take hydration breaks.	High Temperature	Water bottle	Avoid long activity periods without water.	Moderate
3	Sun Protection	Apply SPF 30 sunscreen before outdoor exposure.	Moderate UV	SPF 30 sunscreen	Avoid staying in direct sun for long periods.	Moderate
4	Sun Protection	Apply SPF 50+ sunscreen and use protective gear.	High UV	SPF 50+ sunscreen, hat, sunglasses	Avoid prolonged direct sun exposure.	High
5	Rain Protection	Carry an umbrella and wear water-resistant shoes.	Rain	Umbrella, waterproof shoes	Avoid wet or slippery surfaces where possible.	Moderate
6	Winter Protection	Wear a winter coat, gloves, and boots.	Cold/Snow	Winter coat, gloves, boots	Avoid prolonged cold exposure.	Moderate
7	Surface Safety	Be cautious of slippery ground and reduced visibility.	Snow/Ice	Winter boots	Avoid icy walkways and unsafe travel routes.	High
8	Wind Safety	Secure loose items and avoid exposed outdoor areas.	Windy	Secure clothing/items	Avoid open exposed areas during strong winds.	Moderate
9	Air Quality	Limit prolonged outdoor exposure and reduce intense activity.	Poor AQI	Indoor alternative	Avoid strenuous outdoor activity.	High
10	Indoor Alternative	Move activity indoors when conditions	Heavy Rain/Snow, Extreme UV, Poor AQI	Indoor activity option	Avoid unnecessary outdoor exposure.	High

		are unsafe.				
11	Heat Safety	Choose shaded areas and reduce activity intensity.	High Temperature	Shade, water	Avoid direct sun during peak heat.	High
12	General Weather Prep	Check alerts before leaving and prepare based on conditions.	Changing Weather	Weather protection items	Avoid unplanned exposure to unsafe conditions.	Low

Example Relationship Flow

- weather_forecast.weather_code_id = 8
- weather_codes.weather_code_id = 8
- weather_codes.condition_name = Heavy Rain
- activities.activity_name = Museum Visit
- safety_guidance.recommended_item = Umbrella
- Description returned: Heavy Rain -> Museum Visit + Umbrella Guidance

This design prevents duplicate weather descriptions, activity descriptions, and safety guidance from being stored repeatedly in the forecast or recommendation tables.